Examiner only

SECTION A

Answer **all** questions.

1. State why chlorine is used in water treatment. [1]

....

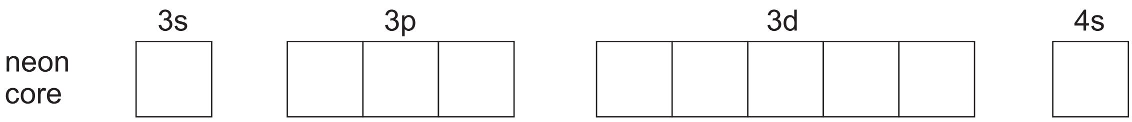
2. State the pattern in electronegativity across Period 3 in the Periodic Table. [1]

....

3. Name the element in Period 3 that has a half-filled set of *p*-orbitals. [1]

....

4. By inserting arrows to represent electrons, complete the electronic structure of a vanadium atom. [1]



5. Magnesium nitrate decomposes when heated to form magnesium oxide, nitrogen dioxide and oxygen.

Balance the equation for this reaction. [1]

$$\ldots\ldots\; Mg(NO_3)_2 \longrightarrow \ldots 2 \ldots\; MgO + \ldots\ldots\; NO_2 + \ldots\ldots\; O_2$$

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6. Consider the following species.

$$^{16}_{8}O^{2-} \qquad ^{18}_{8}O \qquad ^{32}_{16}S^{2-}$$

State whether you agree with the following statements, giving a reason for your conclusion.

(a) $^{32}_{16}S^{2-}$ has twice the number of neutrons that there are in $^{18}_{8}O$. [1]

....

....

(b) The sum of the number of electrons in $^{16}_{8}O^{2-}$ and $^{18}_{8}O$ is equal to the number of electrons in $^{32}_{16}S^{2-}$. [1]

....

....

7. Hydrated copper(II) sulfate has the formula $CuSO_4.5H_2O$.

Calculate the number of moles of oxygen atoms present in 0.5 mol of hydrated copper(II) sulfate. [1]

Oxygen atoms = ... mol

8. A 0.717 g sample of gas occupies 1000 cm^3 at 273 K and 1 atm. Calculate the relative formula mass of the gas. [2]

Relative formula mass = ...

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